

TIDES: A Longitudinal Bilingual Dataset for Modeling Multi-Party Social Dynamics

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Abstract

Group conversations are fundamental to human collaboration, yet standard large language models (LLMs) still struggle with the complexities of multi-party interaction. This challenge persists in part because existing group conversation datasets are often limited to short-term lab settings with contrived tasks, failing to capture the long-term social dynamics of real-world teams. To bridge this gap, we introduce  TIDES, a high-resolution longitudinal dataset tracking 12 university project teams over a full semester. Comprising **76,434** utterances in both English and Korean from in-person meetings, TIDES provides a naturalistic record of teams working on self-managed projects. Our socio-structural annotations—covering interaction types, emergent roles, and development stages—allow for modeling of team evolution over months. Experiments demonstrate that models fine-tuned on TIDES show significant improvements in next-speaker prediction (64.53%, outperforming proprietary models in zero-shot) and, through cross-domain transfer, achieve comparable performance (within 2.1 percentage points) to published state-of-the-art on the AMI Meeting Corpus with approximately 42% less training data. However, preference ratings from human evaluators reveal an *orthogonality between structure and content*: while socio-structural data enhances next-turn prediction, it does not necessarily yield more human-preferred utterances. This suggests that structural and semantic modeling in group dynamics are more decoupled than previously assumed.

1 Introduction

Group conversations are fundamental to human collaboration, yet standard large language models (LLMs), despite their rapid progress in dyadic settings, continue to struggle when deployed in multi-party interactions (Tan et al., 2023; Wei et al., 2023). Unlike dyadic settings, successful group collaboration requires complex social coordination: understanding latent social dynamics (Zhou et al., 2025; Gu et al., 2022) and managing turn-taking across multiple speakers (Ekstedt & Skantze, 2020; Hilgert & Niehues, 2025; Castillo-López et al., 2025). These demands go beyond linguistic proficiency alone, requiring agents to infer and update the latent social dynamics of multiple participants as the interaction unfolds. This challenge is further compounded by the fact that social relationships and team dynamics are not static, but continuously evolve over prolonged interactions (Fang et al., 2025). Developing agents capable of naturalistic multi-party interaction therefore necessitates high-quality, longitudinal resources that capture authentic social dynamics as they naturally unfold.

However, existing multi-party datasets remain insufficient for capturing the social dynamics of collaboration as they unfold in the wild. Many widely-used datasets rely on lab-based observations (Carletta et al., 2005; Karadzhov et al., 2023), scripted media (Poria et al., 2019; Chen et al., 2020; Zhu et al., 2021), or synthesized conversations (Kirstein et al., 2025; Jang et al., 2023), and while such settings can approximate multi-party interaction, they often miss the real stakes, shared history, and evolving interpersonal dependencies that characterize authentic teamwork. These social conditions matter because latent social dynamics—such as influence, alignment, tension, and role occupancy—do not simply appear within a

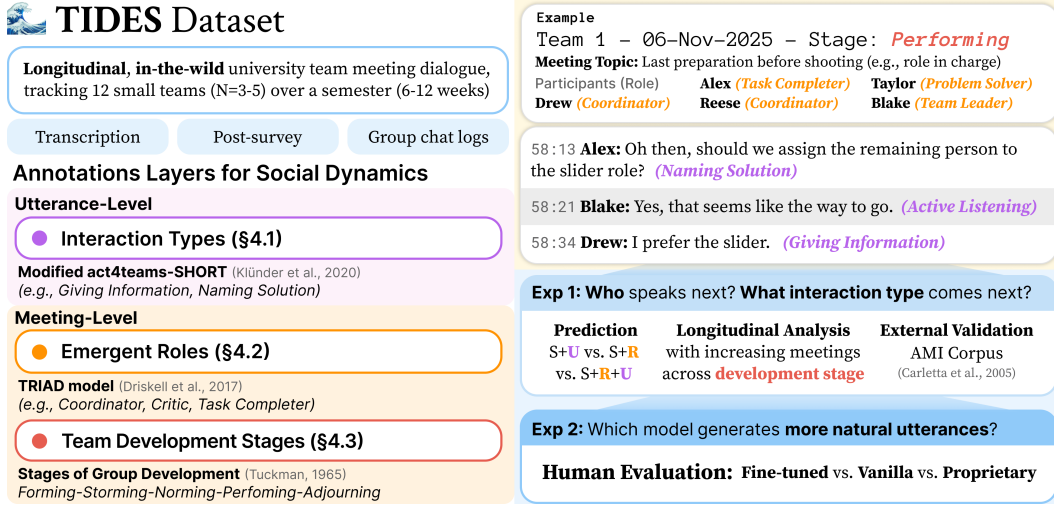


Figure 1: Overview of TIDES. TIDES is a longitudinal, in-the-wild bilingual dataset of university team project meetings, comprising 12 teams, 89 meetings, and 76K+ utterances collected over 6–12 weeks. Beyond transcripts, it provides layered annotations for social dynamics, including utterance-level interaction types, meeting-level emergent roles, and team development stages. The figure also shows an annotated example and the tasks enabled by these annotations, including next-speaker and interaction-type prediction, longitudinal analysis, external validation, and human evaluation for next-utterance generation.

single isolated exchange; they emerge through repeated collaboration as team members negotiate responsibility, expertise, and participation over time. However, even naturalistic meeting corpora (Carletta et al., 2005; Shriberg et al., 2004; Van Segbroeck et al., 2020) mostly capture only short-term episodes, limiting researchers’ abilities to study how such dynamics accumulate and shift across a team’s lifespan. Prior work has also centered on predefined or assigned roles (Jurgens et al., 2023), despite real teams more often exhibiting emergent functional roles that arise from ongoing interaction and changing task demands (Benne & Sheats, 1948).

To begin to bridge this gap, we introduce TIDES, Team Interaction and Dynamic Emergent Social-roles, a longitudinal bilingual dataset tracking real-world university project teams over a full semester. University courses provide a naturalistic setting where teams form organically, collaborate over weeks toward shared goals, and operate under real stakes (i.e., academic grading). We tracked 12 teams over a semester across a diverse range of university classes, recording their actual project meetings to collect a total of 76,434 utterances across 89 meetings (110 hours). To capture not only what teams said but also how their latent social dynamics evolved, participants completed post-meeting surveys after each session, reporting collaboration satisfaction and their perceptions of each member’s emergent role. The resulting dataset pairs privacy-preserved transcripts with longitudinal social annotations, including meeting-level emergent roles, utterance-level interaction types, team progress stages, and silence gaps (annotated examples in Appendix C).

The longitudinal nature of TIDES, in combination with the social annotations, enables us to take steps toward a vision of more socially-aware LLMs for group conversation, addressing four initial questions: (1) *Can models learn turn-taking and interaction patterns from longitudinal team data?* We evaluate next-speaker and next-intention prediction under three context conditions that vary the balance between utterance content and social structure. (2) *How much team-specific data is needed for a model to adequately capture a team’s interaction dynamics?* We train on incrementally more meetings from individual teams, measuring when prediction performance plateaus. (3) *Do these patterns generalize to unseen meeting corpora?* We evaluate on the AMI Meeting Corpus, directly comparing against published state-of-the-art. (4) *Does structural understanding translate to generating utterances that humans*

74 *find natural?* We compare fine-tuned, vanilla, and proprietary models through automatic
75 metrics and a human evaluation on Prolific.

76 Experiments demonstrate that models fine-tuned on TIDES develop a strong understand-
77 ing of turn-taking and interaction patterns. For next-speaker prediction, our fine-tuned
78 model achieves an accuracy of 64.53%, a significant 13.8 percentage point improvement over
79 the baseline. Our data efficiency experiments demonstrate that much of this performance
80 increase is achieved after the first three-to-four team meetings, suggesting that lightweight,
81 application-specific supplements to this data may be feasible to collect in the future. Fur-
82 thermore, through cross-domain transfer to the AMI Corpus (Carletta et al., 2005), our
83 model achieves comparable performance to the published state-of-the-art while utilizing
84 approximately 42% less training data (i.e., utterances). However, preference ratings from hu-
85 man evaluators reveal that a model’s conversational structure prediction accuracy does not
86 seamlessly translate into human-preferred utterance generation. While fine-tuned models
87 outperform baselines in predicting the correct speaker, human judges uniformly favor the
88 vanilla models for their naturalness and coherence across multiple dimensions. This exposes
89 an orthogonality between structure and content: structural modeling (next-turn prediction)
90 and semantic modeling (generating human-preferred content) in group dynamics may be
91 more decoupled than previously assumed.

92 While our experiments primarily evaluate local prediction and generation within short
93 context windows, the longitudinal annotations in TIDES provide the foundation for mod-
94 eling how team dynamics evolve across meetings—a direction that initial evidence in this
95 paper, such as the correlation between team development stage and prediction accuracy
96 (Appendix K), begins to support.

97 2 Related Work

98 **Multi-Party Dialogue Datasets.** Prior work has introduced a range of multi-party meeting
99 and discussion datasets, including the AMI and ICSI Meeting Corpora (Carletta et al.,
100 2005; Shriberg et al., 2004), which provide foundational recordings of real meetings and
101 have supported extensive research on dialogue structure, speaker behavior, and summa-
102 rization. More recent resources such as MeetingBank (Hu et al., 2023), QMSum (Zhong
103 et al., 2021), and ELITR (Nedoluzhko et al., 2022) have expanded the scale and utility of
104 meeting corpora, particularly for summarization and automatic minuting, while datasets
105 like DeliData (Karadzhov et al., 2023) focus on deliberative problem-solving interactions
106 and MPDD (Chen et al., 2020) and FAME (Kirstein et al., 2025) provide controlled or syn-
107 thetic settings for analyzing interpersonal dynamics. However, most existing resources are
108 limited in at least one critical dimension: most capture short-term interactions rather than
109 longitudinal collaboration, some are drawn from scripted or institutional settings that do
110 not fully reflect authentic teamwork, and others rely on assigned, fictional, or synthetic roles
111 rather than emergent team dynamics. In contrast, TIDES captures real-world collaborative
112 behavior *in the wild*, at scale and over a longer time span, enabling the study of how social
113 roles, interaction patterns, and team development evolve throughout the teamwork.

114 **Socially-Aware Dialogue Modeling.** Understanding group dynamics computationally
115 requires more than just dialogue transcripts. It demands annotations of the social structures
116 that shape interaction. Organizational science has long studied how teams develop over
117 time, from Tuckman’s stage model (Tuckman, 1965) to emergent role theories (Kauffeld et al.,
118 2018), yet these frameworks have seen limited adoption in NLP, due to the lack of suitably
119 annotated data. Existing benchmarks for social intelligence rely on dyadic or scripted
120 conversations (Zhou et al., 2024; Zhan et al., 2023; Rashid & Blanco, 2018; Tiginova et al.,
121 2021; Jia et al., 2021; Jurgens et al., 2023), which do not capture the longitudinal, multi-party
122 dynamics that characterize real teamwork. TIDES addresses this with socio-structural
123 annotations—interaction types, emergent roles, and development stages. These annotations
124 enable models to tackle core aspects of group dynamics, such as predicting who speaks next
125 and what type of contribution they make. From a practical standpoint, the longitudinal
126 structure of TIDES raises the question of how much team-specific data is sufficient for
127 models to develop an adequate understanding of a team’s dynamics. Looking beyond our

own corpus, the same annotations also open the question of whether turn-taking patterns learned from TIDES generalize to multi-party conversations in other domains.

Conversational Agents in Group Settings. From an HCI perspective, researchers have investigated how conversational agents can support cooperation and facilitate participation in group discussions (Claggett et al., 2025; Kim et al., 2020; Houde et al., 2025), with a complementary line equipping agents with internal reasoning about *when and why* to speak (Liu et al., 2025). Such approaches assume that richer reasoning about group structure will lead to more natural contributions. However, systematic human evaluation of whether structural understanding actually improves generation quality in multi-party settings is largely absent—existing evaluations typically rely on structural metrics or LLM-based judges. Moreover, these efforts are still typically grounded in dyadic, simulated, short-term settings, leaving a gap in authentic, longitudinal resources. TIDES provides the resources to directly test this assumption. By comparing models that leverage social structure during generation against those that generate without such context, and evaluating both through systematic human judgment, we can assess whether structural understanding and natural generation are complementary—or more orthogonal than commonly assumed.

3 Data Collection

We collected authentic, longitudinal team dynamics by tracking real university project teams throughout a semester. Teams formed organically, determined project goals and milestones within the guidelines of course projects, collaborated under real stakes (i.e., grade evaluation), and interacted repeatedly over weeks toward shared goals. This section describes the recruitment and instrumentation process, the data we collected across meetings, surveys, and messaging logs, and the post-processing steps taken to prepare the data for analysis.

3.1 Collection Procedure

We recruited 12 student teams (total $N = 50$) enrolled in university courses at full-time Korean universities during the Fall 2025 semester. Participants were recruited through a combination of official (university-managed) and unofficial (student-managed) announcement boards and lists. Teams ranged from 3 to 5 members, and their projects ranged in duration from 6 to 12 weeks across a variety of disciplines, including Design, Computer Science, and Industrial Engineering. Prior to the study, all participants attended an orientation session where they were briefed on data collection procedures. Throughout the semester, teams were asked to keep audio recordings of all team project meetings and to submit the recordings to the research team after each session. Following each meeting, every team member individually completed a post-meeting survey so that we could capture meeting-specific perceptions of collaboration and track how team dynamics changed over time. At the end of the semester, each team completed a final team-level survey to reflect on their overall collaboration and project outcome after their last meeting. Additionally, chat message logs regarding the project between team members were collected from all but one team, which did not use a messaging platform during the project. This experiment was approved by our institution’s Institutional Review Board and we compensated each team with 500,000 KRW (approximately 325 USD).

3.2 Post-processing

Converting 110+ hours of bilingual group conversation audio into research-ready transcripts required a six-stage pipeline. We first transcribed all recordings with Whisper Large-V3 (Radford et al., 2023), then assigned speaker labels through a custom diarization pipeline combining pyannote 3.1 (Bredin, 2023; Plaquet & Bredin, 2023) VAD with ECAPA-TDNN (Desplanques et al., 2020), re-clustering to the known team size.

Next, we removed personally identifiable information using Microsoft Presidio (Microsoft, 2023) for the five English-speaking teams and a locally-run Qwen3-30B-A3B (Qwen Team, 2025) for the seven Korean-speaking teams, where no comparable off-the-shelf tool exists.

178 All Korean transcripts were then translated to English by the same local LLM. To resolve
 179 the cross-session speaker identity problem—where the same person may receive different
 180 labels across recordings—we unified speaker labels using WeSpeaker (Wang et al., 2023)
 181 embeddings and the Hungarian algorithm (Kuhn, 1955), producing 422 consistent mappings
 182 across 12 teams. Finally, we refined utterance boundaries with GPT-5-mini guided by
 183 Language Development Project transcription rules (MacWhinney, 2000), yielding 1,238
 184 splits and 309 merges across 105 transcript files.

185 All stages except utterance refinement ran entirely on local machines; the refinement stage
 186 used the OpenAI API on already-anonymized transcripts only. Full pipeline details, model
 187 configurations, and processing statistics are provided in Appendix F.

188 4 Data Annotation

189 After collecting and post-processing the recordings, we constructed multiple annotation
 190 layers to capture social dynamics in small teams. These annotations drew on different
 191 sources: meeting transcripts for utterance-level interaction types, post-surveys for emergent
 192 role assignment, and end-of-semester group reflection for meeting-level team development
 193 stages.

194 4.1 Utterance-level Interaction Type

195 We annotated utterance-level interaction types using a modified version of
 196 act4teams-SHORT (Klunder et al., 2020), which captures interaction behaviors in group
 197 problem-solving at the utterance level. However, because act4teams-SHORT largely omits
 198 the social activity dimension, we extended it to better capture the social signals. Specifically,
 199 drawing from the original act4teams taxonomy (Kauffeld et al., 2018), we separated
 200 *Counterproductivity* into *Social Negative* and *Product Negative*, and readopted *Social/Humor*,
 201 *Active Listening*, and *Others*. This resulted in our modified act4teams-SHORT scheme with a
 202 total of 15 categories.

203 To construct human-labeled gold data, we recruited 260 annotators through Prolific and
 204 assigned three independent annotators to each utterance. We then aggregated labels using
 205 majority voting and retained labels agreed upon by at least two of the three annotators,
 206 resulting in 5,744 human-labeled gold utterances ($\simeq 7.5\%$ of all utterances). We then
 207 fine-tuned gemma-3-12B on the gold data and evaluated several candidate models under a
 208 leave-one-team-out setup, in which one team was held out from fine-tuning to reduce data
 209 leakage across teams. We selected gemma-3-12B for large-scale annotation because it achieved
 210 the highest macro-F1 score (0.54). Considering the complexity of the 15-category taxonomy
 211 and the subjective nature of the task, and moderate human agreement (Fleiss’ $\kappa = 0.400$),
 212 we deemed this performance sufficient for large-scale annotation (Wong et al., 2021), and
 213 used the fine-tuned model to annotate the remaining 70K+ utterances. Additional details
 214 on the crowdsourcing process, model selection, and analysis on gold data are provided in
 215 the Appendix A.

216 4.2 Emergent Roles

217 We assigned one peer-perception-based emergent role to each participant at each meeting.
 218 In the survey, participants evaluated their teammates along three dimensions in the TRIAD
 219 model Driskell et al. (2017)—*Dominance*, *Sociability*, and *Task Orientation*—using nine 5-point
 220 Likert items (three items per dimension). We chose this dimension-based design rather than
 221 asking participants to directly assign one of the 13 TRIAD roles, as direct role categorization
 222 is difficult for non-expert participants. Detailed survey items are provided in Appendix E.

223 Because our survey used a 5-point Likert scale whereas TRIAD roles are defined in a
 224 different coordinate space (7-point Likert scale), we normalized the survey scores before
 225 role assignment. We used team-specific cumulative normalization, where each participant’s
 226 score at meeting m was standardized using the responses from the same team observed up

227 to and including that meeting:

$$z_{t,m,i}^{(k)} = \frac{x_{t,m,i}^{(k)} - \mu_{t,\leq m}^{(k)}}{\sigma_{t,\leq m}^{(k)}}. \quad (1)$$

228 Here, $x_{t,m,i}^{(k)}$ denotes participant i 's average survey score on dimension k in team t at meeting
 229 m , and $\mu_{t,\leq m}^{(k)}$ and $\sigma_{t,\leq m}^{(k)}$ denote the mean and standard deviation of that dimension computed
 230 from all responses in team t up to meeting m . We then assigned the role whose standardized
 231 prototype $\tilde{\mathbf{p}}_r$ was closest in Euclidean distance:

$$\hat{r}_{t,m,i} = \arg \min_{r \in \mathcal{R}} \|\mathbf{z}_{t,m,i} - \tilde{\mathbf{p}}_r\|_2. \quad (2)$$

232 $\hat{r}_{t,m,i}$ denotes the emergent role assigned to participant i in team t at meeting m . This
 233 procedure allowed us to assign an emergent role to each participant at each meeting while
 234 preserving team-specific temporal context. To support alternative analyses beyond discrete
 235 role labels, TIDES also includes the raw scores on the three dimensions for each participant
 236 at each meeting.

237 4.3 Team Development Stage Annotation

238 To annotate how teams evolved over time, we assigned each meeting a stage from Tuckman's
 239 team development model: Forming, Storming, Norming, Performing, or Adjourning (Tuck-
 240 man, 1965). During the final post-data collection meeting, team members gathered and
 241 reviewed previous meetings prepared by the research team and discussed which develop-
 242 ment stage best characterized the team stage of each meeting. The agreed-upon stage was
 243 then recorded as a meeting-level label.

244 4.4 Annotated Dataset Summary

245 After post-processing and annotation, the dataset contains 76,434 utterances collected from
 246 89 meetings across 12 student teams ($N = 50$). Team size ranged from 3 to 5 members: 2
 247 teams had 3 members, 6 teams had 4 members, and 4 teams had 5 members. 7 teams primar-
 248 ily communicated in Korean and 5 teams in English. Using post-meeting peer-perception
 249 surveys, we additionally assigned 352 meeting-level emergent role labels, corresponding to
 250 one role per participant per meeting where survey responses were available.

251 The annotated interaction types were skewed toward a small number of frequent collabora-
 252 tive behaviors. *Giving Information* was the most common category (34.92%), followed by
 253 *Linking Solutions* (12.86%) and *Active Listening* (11.73%), suggesting that the meetings were
 254 dominated by task-relevant information sharing, solution-building, and responsive discus-
 255 sion. Less frequent categories such as *Task/Process Negative* (0.95%), *Social Negative* (0.80%),
 256 and *Linking & Connecting* (0.07%) appeared only rarely. For emergent roles, *Problem Solver*
 257 (17.61%), *Coordinator* (16.76%), and *Critic* (15.91%) were the most common, indicating that
 258 task-oriented and coordination-oriented roles appeared more often than socially disruptive
 259 or off-task roles. Detailed team-level statistics and full label distributions are provided in
 260 Appendix B; annotated transcript excerpts are shown in Appendix C.

261 5 Experiments

262 We evaluate TIDES through two experiments: (1) next-speaker and intention prediction,
 263 with follow-up analyses on data efficiency and external validation on the AMI Corpus, and
 264 (2) next-utterance generation with human evaluation.

265 5.1 Experimental Setup

266 **Common configuration.** Unless stated otherwise, we fine-tune with LoRA (rank 16, $\alpha = 32$)
 267 on Gemma-3-12B-IT, trained on 32,243 samples (Teams 1–5, 8–12), validated on 1,857 samples

(Team 4), and tested on 5,094 samples (Teams 6&7) with 5-turn context windows. Fine-tuned models use completion-only loss; inference uses single-token logit scoring for open-source models and generative decoding (temperature 0) for proprietary APIs. All results are single runs. We additionally validate with Llama-3.1-8B-Instruct and benchmark four proprietary models (GPT-5.4, GPT-5.4-mini, Opus 4.6, Sonnet 4.6) in zero-shot. All Korean transcripts were translated to English during post-processing (§3.2); language labels (Korean/English) throughout refer to the language originally spoken during meetings.

We design two experiments:

1. **Prediction and analysis.** To probe whether models can learn group-level turn-taking patterns from TIDES, we evaluate next-speaker prediction (3–5 classes per team) and next-intention prediction (14 classes) under three context conditions: **SU** (Speaker + Utterance), **SRU** (+ annotated role and Tuckman stage), and **S+R** (structure only—speaker IDs, roles, and stages, with all utterance text removed). To investigate how much team-specific data is required to achieve adequate prediction performance—a practical consideration for deploying such models to new teams—we further analyze *data efficiency* by training single-team models with chronologically increasing meetings on Team 10 (Korean, 4 members, 7 meetings) and Team 5 (English, 5 members, 10 meetings). To test whether TIDES-trained models generalize beyond our corpus, we conduct *external validation* on the AMI Meeting Corpus (Carletta et al., 2005) (138 meetings, 4 English speakers each) using Llama-3.1-8B for direct comparison with Hilgert & Niehues (2025), who report 47.85% using AMI plus MultiLIGHT data.
2. **Generation.** To examine whether structural understanding translates to realistic utterance generation, we compare **FT-A** (plain generation: “SPEAKER: utterance”), **FT-B** (social cue reasoning: the model generates an explicit role/intention prediction before producing the utterance), and **V-A** (vanilla Gemma-3-12B, no fine-tuning), plus four proprietary models. Each generates continuations at 1, 3, and 5 turns (100 samples per length). We evaluate with automatic metrics (speaker accuracy, semantic similarity), a human evaluation on Prolific ($N=66$ evaluators, 1,142 approved pairwise judgments, £4.50/evaluator), and LLM-as-judge evaluation.

5.2 Results

Experiment 1a: Speaker prediction. As a reference point, the bigram baseline—which predicts the most frequent next speaker given the current speaker—achieves 50.75%. Fine-tuning on TIDES yields large gains (Table 1): Gemma FT-SU reaches 64.53%, +13.8 pp over the bigram baseline. Adding role labels helps minimally (FT-SRU 64.82%, $\Delta = +0.3$ pp), but removing text entirely still matches performance (FT-S+R 65.74%), indicating that structural features alone are sufficient for predicting who speaks next. This may reflect that utterance text introduces team-specific lexical patterns (e.g. project topics, jargon) that do not transfer to held-out teams, whereas structural features—speaker order, annotated roles, and development stages—capture more generalizable turn-taking regularities. Fine-tuning outperforms all proprietary zero-shot models, with Gemma FT-SU (64.53%) exceeding Opus 4.6 (63.25%) and GPT-5.4 (61.97%). Llama-3.1-8B replicates all patterns with comparable accuracy (Appendix G). A longitudinal signal is also evident. Speaker prediction accuracy increases from 57.8% in Forming-stage meetings to 70.3% in Performing-stage meetings (+12.5 pp; Appendix K), consistent with the theoretical expectation that established teams develop more predictable patterns.

Experiment 1b: Intention prediction. Intention prediction is a substantially harder task, likely due both to the larger number of classes and the subjective nature of categorization (Table 1). The best model (FT-SU, 32.96%) only marginally exceeds the majority baseline (30.05%), and unlike speaker prediction, text is necessary (FT-S+R drops to 31.17%). No zero-shot model—proprietary or open-source—surpasses the majority baseline, reflecting inherent task ambiguity ($\kappa = 0.339$).

Experiment 1c: Data efficiency. Extending the prediction analysis to single-team settings, both teams show rapid improvement that plateaus after 3–4 meetings (Table 2). Team 10

Model	Context	Type	Speaker		Intention	
			Acc.	Δ Bigram	Acc.	Δ Maj.
<i>Baselines</i>						
Random	—	—	30.22	—	7.14	—
Bigram	—	—	50.75	(ref)	—	—
Majority	—	—	—	—	30.05	(ref)
<i>Zero-shot / Vanilla</i>						
Gemma-3-12B	SU	vanilla	43.56	−7.2	25.48	−4.6
GPT-5.4	SU	zero-shot	61.97	+11.2	25.68	−4.4
Opus 4.6	SU	zero-shot	63.25	+12.5	26.68	−3.4
<i>Fine-tuned (Gemma-3-12B-IT + LoRA)</i>						
Gemma-3-12B	SU	FT	64.53	+13.8	32.96	+2.9
Gemma-3-12B	SRU	FT	64.82	+14.1	32.53	+2.5
Gemma-3-12B	S+R	FT	65.74	+15.0	31.17	+1.1

Full results including Llama-3.1-8B, additional proprietary models, and few-shot baselines in Appendix G. Majority baseline for intention = “Giving Information” (30.05%).

Table 1: Next-speaker and next-intention prediction on the TIDES test set (5,094 samples, Teams 6&7). Context conditions: SU = Speaker + Utterance; SRU = + Role + Tuckman stage; S+R = structure only (no text). **Bold** = best. *Italic* = proprietary zero-shot.

		Gemma-3-12B		Llama-3.1-8B		Method	Train Data	Acc.
Mtgs	Train N	Acc.	% ceil.	Acc.	% ceil.			
Team 10 (Korean, 4 members, 7 meetings)						Baselines (H&N)		
						Llama 8B zero-shot	—	34.88
						Llama 70B zero-shot	—	35.81
1	276	50.87%	75.8%	53.61%	75.6%	Ours (Llama-3.1-8B + LoRA)		
3	795	55.37%	82.5%	58.46%	82.4%	TIDES-FT	TIDES 32K	35.33
4	1,073	62.62%	93.3%	63.20%	89.2%	TIDES-unmrg	TIDES 62K	40.04
6	1,811	65.82%	98.1%	64.35%	90.8%	TIDES+AMI bal.	124K	45.79
Team 5 (English, 5 members, 10 meetings)						Published SOTA (H&N)		
1	203	50.00%	73.2%	60.73%	99.6%	Llama 8B FT	AMI+ML	47.85
3	956	63.84%	93.5%	64.83%	106.3%	H&N = Hilgert & Niehues (2025). ML = MultiLIGHT. TIDES+AMI uses ~42% less		
5	1,648	64.81%	94.9%	65.10%	106.8%			
7	2,570	64.59%	94.6%	64.68%	106.1%			

Ceilings—Gemma: 67.09% (T10), 68.27% (T5); Llama: 70.89% (T10), 60.96% (T5). Llama T5 exceeds ceiling due to small test set (187 samples).

Table 2: Learning curve: single-team fine-tuning with increasing meetings. % of ceiling is relative to each model’s 9-team general model.

H&N = Hilgert & Niehues (2025). ML = MultiLIGHT. TIDES+AMI uses ~42% less data than H&N.

Table 3: External validation on the AMI Meeting Corpus (12,515 test samples, context window of 8). All models use Llama-3.1-8B for direct comparison with Hilgert & Niehues (2025).

reaches 55.37% with just 3 meetings, already 82.5% of the 9-team general model (67.09%). Team 5 reaches 63.84% at 3 meetings (93.5% of ceiling). Both Gemma and Llama show the same rapid convergence, indicating that a few hours of team-specific data can approach the performance of a model trained on the full multi-team corpus.

Experiment 1d: External validation. To test generalization beyond TIDES, we evaluate on AMI. A balanced TIDES+AMI mix reaches **45.79%**, within 2.06 pp of published SOTA with ~42% less training data and cross-domain TIDES substituting for in-domain MultiLIGHT (Table 3). The zero-shot TIDES-FT model (35.33%) matches Hilgert & Niehues’ zero-shot (34.88%), suggesting that TIDES training captures transferable turn-taking patterns. However, the full unbalanced mix (170K, trained for 1 epoch) degrades to 30.02%, suggesting that data balance matters more than volume for cross-corpus transfer.

Experiment 2a: Generation: automatic metrics. FT-B achieves the highest 1-turn speaker accuracy (57.0% vs. FT-A 49.0%, V-A 14.0%), and both FT conditions produce far fewer

Pair	N	Pairwise Preference				Nat.		Coh.		Spkr Con.	
		A%	B%	Tie%	Sig.	A	B	A	B	A	B
FT-A vs FT-B	388	21.4	25.3	53.4		3.21	3.42	3.08	3.40***	3.13	3.41***
FT-A vs V-A	407	9.3	21.1	69.5	***	3.07	3.99***	2.78	4.10***	3.05	3.93***
FT-B vs V-A	347	12.7	21.9	65.4	**	3.14	3.98***	2.82	4.06***	3.04	3.85***

A, B = first, second model in each pair. Nat. = Naturalness, Coh. = Coherence, Spkr Con. = Speaker Consistency (Likert 1–5). Significance: *** $p < .001$, ** $p < .01$ (Bonferroni-corrected).

Table 4: Prolific human evaluation ($N=66$ evaluators, 1,142 approved judgments, £4.50/evaluator). FT-A = plain generation; FT-B = social cue reasoning; V-A = vanilla Gemma-3-12B.

invalid speakers than vanilla (152–178 vs. 786 at 10 turns), suggesting that fine-tuning on TIDES improves structural coherence in generation. However, proprietary models outperform all fine-tuned models: Opus 4.6 reaches 62.0% at 1-turn with zero invalid speakers, and degrades more slowly (39.0% at 5-turn vs. FT-B’s 29.2%). Proprietary models also show higher semantic similarity to ground truth (0.36–0.44 vs. 0.28–0.36; Table 19). Unlike prediction, where fine-tuning closes the gap with proprietary models, generation quality appears to benefit more from model scale.

Experiment 2b: Generation: human evaluation. Despite the results above, human evaluators reveal a contrasting pattern: they **significantly prefer vanilla over fine-tuned outputs** (Table 4). FT-A wins only 9.3% vs. V-A’s 21.1% ($p < .001$); FT-B wins 12.7% vs. 21.9% ($p=.004$). V-A scores 0.8–1.3 Likert points higher on naturalness, coherence, and speaker consistency ($d = 0.46$ – 0.72 ; all p -values remain significant after Bonferroni correction). FT outputs, while contextually grounded, exhibit surface artifacts (truncation, missing punctuation), whereas V-A generates polished but often off-topic continuations. One might attribute this preference gap to these surface differences, but we applied post-processing to all FT outputs before evaluation—correcting capitalization, punctuation, and truncated endings (Appendix L)—yet the gap persisted. This suggests that the difference runs deeper than formatting: **structural and semantic modeling in group dialogue are more decoupled than commonly assumed**—predicting *who* speaks next does not entail generating text that humans judge as realistic. This finding is further complicated by the choice of evaluator: GPT-5.4 mirrors the human preference direction (V-A preferred 72.5%), while Opus 4.6 favors FT-A at 1-turn (72.5%) but reverses at 5-turn (V-A 67.5%; Appendix L.1).

6 Conclusion

We introduced TIDES, a longitudinal bilingual dataset tracking 12 university project teams over a full semester, with socio-structural annotations including emergent roles, interaction types, and team development stages. Through a series of experiments, we demonstrated that fine-tuning on TIDES substantially improves next-speaker prediction—outperforming proprietary large-scale models—and that only a few hours of well-structured meeting data is sufficient for models to capture a team’s interaction dynamics, with learned patterns transferring competitively to the AMI Meeting Corpus.

At the same time, our human evaluation revealed an orthogonality between prediction accuracy and generation quality. Models that excelled at predicting who speaks next did not necessarily generate utterances that humans judged as natural.

Our experiments primarily evaluated local prediction within short context windows. Modeling how team dynamics accumulate and shift across a team’s full lifespan—for example, predicting role transitions, detecting phase shifts in real time, or adapting to evolving team norms—remains an important open direction. The longitudinal structure and social annotations in TIDES open the door to such research, offering a resource where cross-meeting dynamics, role trajectories, and team development can be studied as they naturally unfold.

Ethics Statement

This study was conducted under approval from our institute’s Institutional Review Board (IRB). All participants provided informed consent prior to enrollment and were fully briefed on data collection procedures, including audio recording, during an orientation session. Participation was voluntary, and participants were free to withdraw at any time.

To protect participant privacy, only transcripts are made publicly available. Prior to any analysis or release, transcripts were processed through a multi-stage anonymization pipeline. Personally identifiable information—names, locations, and organizational references—was removed using Microsoft Presidio for English-speaking teams and a locally run Qwen3-30B-A3B model for Korean-speaking teams. All pseudonyms are gender-neutral. We compensated each team 500,000 KRW (approximately 325 USD) for their participation.

For annotation on Prolific, annotators were compensated at 8 GBP (approximately 11 USD) per task (100 utterances), and for human evaluators, each evaluator was compensated at 4.5 GBP (approximately 6 USD) per task. Both rates exceeded the platform’s recommended rate. There was no discrimination in the recruitment of annotators based on any demographic characteristics.

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602 A Details for Utterance-level Interaction Type Annotation

603 A.1 Construction of Human-labeled Gold Data via Prolific

604 For utterance-level interaction type annotation, we recruited 260 annotators through the
 605 crowdsourcing platform Prolific. Annotators accessed our custom-built annotation interface
 606 (Fig. 2), where they first provided their Prolific ID and then proceeded to the task. Before
 607 starting the main annotation task, annotators were instructed to carefully review the defini-
 608 tions of the 15 categories in our modified act4teams-SHORT scheme. To ensure sufficient
 609 understanding of the coding scheme, annotators were required to complete a tutorial quiz
 610 and answer all questions correctly before proceeding. Each annotator was assigned a total of
 611 100 utterances, organized into 10 windows of 10 target utterances each. To help annotators
 612 interpret each utterance in context, we additionally provided the 10 preceding utterances
 613 and 10 following utterances for each window. After completing all 10 windows, annotators
 614 received approximately 11 USD in compensation, subject to a quality check by the research
 615 team.

616 A.2 Interaction Types in Human-labeled Gold Data

617 Table 5 summarizes the distribution of utterance-level interaction types in the human-
 618 labeled gold data. The distribution is imbalanced, with *Giving Information* and *Active*
 619 *Listening* accounting for a large proportion of the annotations, while several categories
 620 such as *Social Negative*, *Task/Process Negative*, and *Linking & Connecting* appear relatively
 621 infrequently.

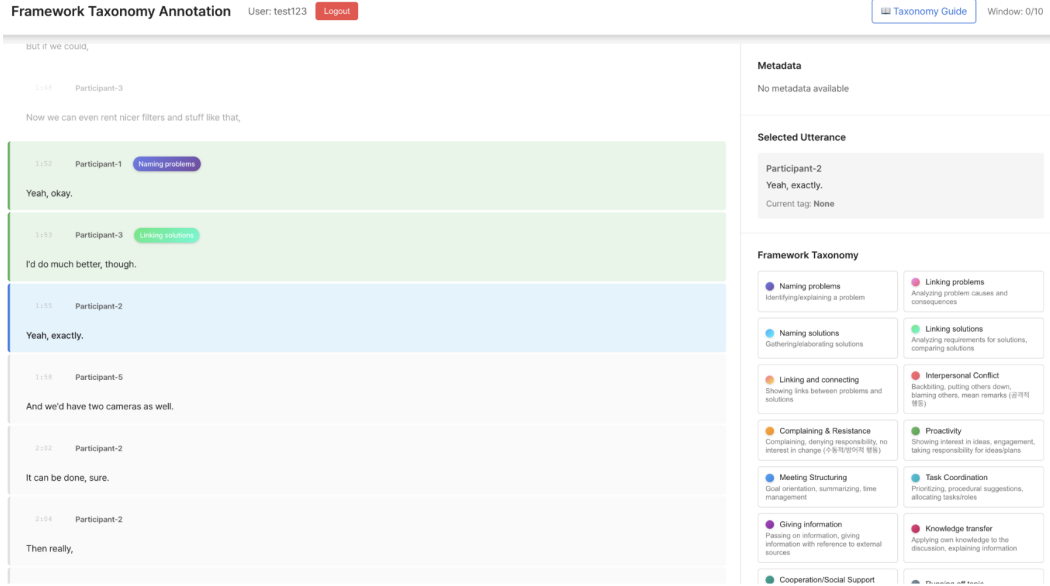


Figure 2: Custom annotation interface used for utterance-level interaction type annotation on Prolific. Annotators reviewed each target utterance with surrounding conversational context and selected one of the 15 modified act4teams-SHORT categories.

Interaction Type	Count	Ratio (%)
Giving Information	1,885	32.82
Active Listening	958	16.68
Linking Solutions	397	6.91
Other / Neutral	385	6.70
Structuring	362	6.30
Naming Solutions	340	5.92
Social / Humor	299	5.21
Proactivity	244	4.25
Naming Problems	222	3.86
Knowledge Transfer	170	2.96
Cooperation	165	2.87
Linking Problems	135	2.35
Social Negative	68	1.18
Task/Process Negative	63	1.10
Linking & Connecting	51	0.89

Table 5: Distribution of utterance-level interaction types in the human-labeled gold data.

622 A.3 Utterance-Level Interaction Type Annotation

623 We compared several base and fine-tuned models on the utterance-level interaction clas-
 624 sification task using a leave-one-team-out validation setup (Table 6). Although fine-
 625 tuned Qwen3-14B achieved the highest accuracy and Cohen’s κ , we selected the fine-tuned
 626 Gemma-3-12B for large-scale annotation because it achieved the best macro-F1 score (0.540),
 627 which we considered the most appropriate primary metric for our imbalanced multi-class
 628 setting. The Fleiss’ κ among human annotators was 0.400, indicating moderate agreement
 629 on this challenging 15-class annotation task.

Model	Acc.	κ	F1
Llama3.1-8B (Base)	0.382	0.286	0.231
Llama3.1-8B (FT)	0.687	<u>0.606</u>	<u>0.528</u>
Qwen3-14B (Base)	0.488	0.397	0.313
Qwen3-14B (FT)	0.691	0.612	0.527
Gemma-3-12B (Base)	0.570	0.477	0.367
Gemma-3-12B (FT)	<u>0.687</u>	0.605	0.540

Table 6: Mean validation performance for utterance-level interaction type classification (leave-one-team-out). Best is bold; second-best is underlined.

(a) Utterance-level interaction type distribution			(b) Emergent role distribution		
Utterance Type	Count	Ratio (%)	Role	Count	Ratio (%)
Giving Information	26,688	34.92	Problem Solver	62	17.61
Linking Solutions	9,826	12.86	Coordinator	59	16.76
Active Listening	8,966	11.73	Critic	56	15.91
Naming Solutions	5,600	7.33	Negative	38	10.80
Structuring	4,834	6.32	Task Motivator	32	9.09
Other / Neutral	4,031	5.27	Team Leader	32	9.09
Proactivity	3,962	5.18	Evaluator	16	4.55
Naming Problems	3,084	4.03	Attention Seeker	13	3.69
Linking Problems	2,550	3.34	Task Completer	11	3.13
Social / Humor	2,503	3.27	Teamwork Support	11	3.13
Cooperation	1,552	2.03	Follower	10	2.84
Knowledge Transfer	1,446	1.89	Power Seeker	7	1.99
Task/Process Negative	724	0.95	Social	5	1.42
Social Negative	611	0.80			
Linking & Connecting	57	0.07			
Total	76,434	100.00	Total	352	100.00

Table 7: Distribution of utterance-level interaction types and emergent roles in the dataset.

B Annotated Dataset Statistics

The amount of conversation varied substantially across teams, ranging from 2,788 to 14,058 utterances per team, with an average of 6,370 utterances per team.

C Transcript Examples

Tables 9 and 10 show representative excerpts illustrating the full annotation layers of TIDES: pseudonymized speaker identity, emergent role, interaction type, and team development stage (in caption). Table 9 additionally shows Korean original utterances alongside their English translations.

D Meeting Metadata and Survey Examples

Table 11 shows the meeting metadata for Team 5, illustrating how team development stages progress over the semester. Table 12 shows post-meeting satisfaction scores and peer-evaluated emergent roles for the same team’s first meeting.

E Full Post-Survey Question Set

To capture the evolution of group dynamics throughout the project, we administered a post-survey immediately after each meeting, as well as a final survey at the end of the project. The

Team	# Utterances	# Meetings
Team_3	14,058	8
Team_9	11,868	7
Team_2	11,631	13
Team_7	6,594	5
Team_5	6,169	10
Team_10	4,488	7
Team_12	4,293	13
Team_1	3,861	5
Team_11	3,859	6
Team_6	3,448	6
Team_8	3,377	5
Team_4	2,788	4
Total	76,434	89

Table 8: Team-level statistics of the dataset.

Speaker	Role	Korean (PII-removed)	English (translated)	Type
Alex (A)	Task Compl.	아 그러면 이제 두 분 중에 남는 분을 슬라이터로 맡기면 되는 건가요?	Oh then, should we assign the remaining person to the slider role?	Naming Sol.
Blake (E)	Team Leader	네, 그래야 할 것 같아요.	Yes, that seems like the way to go.	Active List.
Alex (A)	Task Compl.	저야 당연히 그러면 좋습니니다.	I'd be happy with that, of course.	Cooperation
Blake (E)	Team Leader	아니면 Drew님은 어떤 거 하는 게 편하세요?	Or, Drew, which one would be more convenient for you?	Structuring
Drew (C)	Coordinator	어 저는 슬라이터가 더 좋아요.	I prefer the slider.	Giving Info.
Blake (E)	Team Leader	그러면 맡길까요?	Then shall we assign it to you?	Proactivity

Table 9: Transcript excerpt from Team 1 (Korean → English, meeting 2025-11-06). Team development stage: *Performing*. The team is assigning filming roles for a video production project.

Speaker	Role	Utterance	Type
Riley (E)	Critic	I guess there is some random Academic Institute branch of the node pixel.	Giving Info.
Jordan (C)	Critic	Maybe you should make a copy of our stuff.	Naming Sol.
Morgan (A)	Negative	The LED is so bright that it can be fixed, but the hole is made here and there.	Linking Sol.
Morgan (A)	Negative	But to put several LEDs in a row,	Linking Sol.
Dakota (D)	Prob. Solver	I can't make the boards bend.	Naming Prob.
Dakota (D)	Prob. Solver	I think it's better to use glue gun or bond to connect them with LED lines.	Naming Sol.

Table 10: Transcript excerpt from Team 5 (English, meeting 2025-11-18). Team development stage: *Norming*. The team is building a physical LED prototype for an HCI course project.

645 post-survey consisted of three parts. First, participants provided a brief reflection summa-
646 rizing important decisions, turning points, or notable events from the meeting. Second, they
647 rated their satisfaction with team collaboration using questionnaire items adapted from prior

Date	Stage	Topic (summary)
2025-10-28	Forming	Brainstormed 5 ideas, narrowed to 3
2025-11-03	Forming	Finalized topic, planned implementation
2025-11-10	Storming	Decided on circuits over Arduino
2025-11-18	Norming	Realized neopixel implementation difficulty
2025-12-02	Performing	Redistributed work after rack assembly
2025-12-06	Performing	Bug fixing and integration
2025-12-10	Performing	Button working, display planning
2025-12-12	Performing	Product finalized and functional
2025-12-13	Adjourning	Filmed demo video
2025-12-15	Adjourning	Presentation role division

Table 11: Meeting metadata for Team 5 (English, 5 members). Stages were assigned retrospectively by team consensus via Tuckman self-assessment.

Satisfaction Item	P1	P2	P3	P4	P5	Avg
Clear collaborative patterns	3	3	3	4	5	3.6
Members know their roles	3	5	5	5	5	4.6
Clear goals and working norms	4	5	5	5	5	4.8
Timely responses	4	5	4	5	5	4.6
Frequent communication	5	5	4	5	5	4.8

Member	G1 (Task)	G2 (Social)	G3 (Domin.)	Assigned Role
Morgan (A)	3.58	4.58	4.58	Teamwork Support
Casey (B)	4.33	4.89	4.89	Coordinator
Jordan (C)	4.44	4.11	4.22	Power Seeker
Dakota (D)	3.78	4.78	4.56	Problem Solver
Riley (E)	3.67	4.33	4.11	Critic

Table 12: Post-meeting satisfaction survey and emergent roles for Team 5 (meeting 2025-10-28, stage: *Forming*). Satisfaction items are rated 1–5; roles are derived from peer-evaluation scores across three TRIAD dimensions.

work on collaboration and team processes [Ku et al. \(2013\)](#). Third, to capture emergent roles within the team, participants evaluated each of their teammates using nine peer-assessment items. These items reflect the three dimensions of the TRIAD model—Dominance, Sociability, and Task Orientation—with three questions for each dimension ([Driskell et al., 2017](#)). At the end of the semester, we additionally administered a final survey to assess participants’ overall satisfaction and perceived quality of the project outcome. We also asked each team to collaboratively reflect on their meeting history and classify each meeting according to Tuckman’s Team Development Model—Forming, Storming, Norming, Performing, and Adjourning—along with a rationale for each classification ([Tuckman, 1965](#)).

Below, we provide the full set of survey questions used in our study.

E.1 Daily Meeting Reflection

- Please share the important decision or turning point during the conversation of today’s meeting.

E.2 Team Collaboration Satisfaction

Note: All items are measured on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree).

1. My team develops clear collaborative patterns to increase team learning efficiency.
2. My team members clearly know their roles during the collaboration.
3. My team has an efficient way to track the edition of documents.

- 666 4. My team sets clear goals and establishes working norms.
- 667 5. My team members reply to all responses in a timely manner.
- 668 6. My team members communicate with each other frequently.
- 669 7. I trust each team member can complete his/her work on time.
- 670 8. My team is receiving feedback from each other.
- 671 9. Communicating with team members regularly helps me to understand the team project better.
- 672
- 673 10. My team members encourage open communication with each other.
- 674 11. My team members communicate in a courteous tone.

675 **E.3 Emergent Role Survey (Peer Evaluation)**

676 *Note: These questions evaluate the behavior of each team member on a 5-point frequency scale (1 =*
 677 *Never to 5 = Always).*

- 678 1. Did this team member actively lead group discussions?
- 679 2. Did this team member clearly present the team’s activities or direction?
- 680 3. Did this team member assert their opinions in decision-making and influence the team?
- 681 4. Did this team member create a positive and comfortable atmosphere in the team?
- 682 5. Did this team member respect and support other team members’ feelings and opinions?
- 683 6. Did this team member help mediate or facilitate smooth communication during conflicts?
- 684 7. Did this team member help the team stay focused on its shared goals and avoid distractions?
- 685
- 686 8. Did this team member fulfill their assigned role responsibly and on time?
- 687 9. Did this team member show a diligent and meticulous attitude to improve task quality?

688 **E.4 Final Survey (Individual Assessment)**

- 689 1. How satisfied are you overall with the project results?
- 690 2. Do you think the project has achieved its intended outcome?
- 691 3. How do you rate the quality of the project results?

692 **E.5 Final Survey (Team Reflection)**

- 693 • **Group Discussion Task:** Looking back at all previous meetings, discuss with your
 694 team members and assign each meeting to the corresponding stage of Tuckman’s Team
 695 Development Model (Forming, Storming, Norming, Performing, Adjourning). Please
 696 provide the rationale for your classifications.

697 **F Post-processing Pipeline Details**

698 This appendix provides full technical details for the six-stage post-processing pipeline
 699 described in Section 3.2. The pipeline converts raw audio into privacy-preserved, speaker-
 700 identified transcripts through the following stages: (1) ASR, (2) speaker diarization, (3) PII
 701 removal, (4) translation, (5) cross-session speaker unification, and (6) utterance boundary
 702 refinement.

703 **F.1 Stage 1: ASR**

704 We transcribed all audio files using Whisper Large-V3 (Radford et al., 2023) with float16
 705 precision on a single RTX A6000 GPU. Table 13 summarizes the configuration. Au-
 706 dio files were first preprocessed with FFmpeg to remove silence regions using the filter

707 silenceremove=1:0:-50dB and resampled to 16 kHz mono. We loaded Whisper Large-V3
 708 via the HuggingFace transformers pipeline with hallucination reduction enabled through a
 709 repetition penalty of 1.1 and no_repeat_ngram_size of 3.

Parameter	Value
Model	openai/whisper-large-v3
Device / Precision	CUDA / float16
Batch size	24
Temperature	0.2
Entropy threshold	2.8
Sample rate	16,000 Hz
Repetition penalty	1.1
No-repeat n-gram size	3
Return timestamps	Segment-level

Table 13: ASR configuration for Whisper Large-V3.

710 F.2 Stage 2: Speaker Diarization

711 Raw Whisper output does not distinguish speakers. We developed a *Smart Pipeline* that
 712 combines pyannote 3.1 (Bredin, 2023; Plaquet & Bredin, 2023) for voice activity detection
 713 with ECAPA-TDNN (Desplanques et al., 2020) speaker embeddings (192-dim) and Median
 714 Absolute Deviation outlier filtering, then re-clusters to the known team size N , ensuring
 715 correct speaker counts by construction.

716 *Smart Pipeline.* The pipeline proceeds in five steps:

- 717 1. **VAD:** pyannote 3.1 (pyannote/speaker-diarization-3.1) segments the audio into speech
 718 and non-speech regions.
- 719 2. **Segment extraction:** Speech regions are extracted as individual audio segments.
- 720 3. **Embedding extraction:** Each segment is encoded with ECAPA-TDNN
 721 (speechbrain/spkrec-ecapa-voxceleb), producing a 192-dimensional speaker em-
 722 bedding. Segments shorter than 0.5 s are zero-padded.
- 723 4. **MAD filtering:** Speaker-level embeddings are computed by averaging segment embed-
 724 dings per speaker, with Median Absolute Deviation filtering to remove outlier segments.
- 725 5. **Re-clustering:** Embeddings are re-clustered to the known team size N using multi-
 726 ple methods (KMeans, Agglomerative with ward/complete/average linkage, Spectral
 727 clustering), and the best result is selected.

728 F.3 Stage 3: PII Removal

729 English and Korean require fundamentally different PII removal approaches. For English,
 730 we used Microsoft Presidio (Microsoft, 2023), a rule-based detection framework with entity
 731 linking and consistent pseudonymization. For Korean, we employed Qwen3-30B-A3B-
 732 Instruct-2507 (Qwen Team, 2025) running locally as a context-aware detector; an earlier
 733 regex-based approach had yielded a 91% false-positive rate, which the LLM-based pipeline
 734 substantially reduced, though 375 manual corrections were still needed. All replacements
 735 use gender-neutral pseudonyms (e.g., Alex, Jordan, Taylor).

736 **English Pipeline.** For the five English/mixed teams (329,537 segments), we used Microsoft
 737 Presidio with three custom components:

- 738 • **EntityLinker:** Maps name variants (e.g., “Mary,” “MJ,” “Jane”) to canonical forms.
- 739 • **ConsistentAnonymizer:** Ensures the same entity always maps to the same pseudonym
 740 across the entire dataset.

- **PresidioPIIMaskerV2:** Wraps the Presidio AnalyzerEngine with custom Korean phone number and ID recognizers.

Detected entity types include: PERSON, LOCATION, ORGANIZATION, EMAIL, PHONE_NUMBER, URL, and DATE_TIME.

Korean Pipeline. For the seven Korean teams (48,697 segments), we employed Qwen3-30B-A3B-Instruct-2507 running locally (bfloat16, device_map=auto) in a three-stage process:

1. **Detection:** The LLM identifies PERSON, LOCATION, and ORGANIZATION entities directly from Korean text, generating an entity_mapping.json.
2. **Replacement:** Detected entities are replaced using regex patterns with Korean-aware word boundary matching (lookbehind for Hangul/ASCII, lookahead for ASCII only, since Korean particles attach directly to names).
3. **Translation:** PII-replaced Korean text is translated to English by the same Qwen3 model, running as a subprocess to ensure GPU memory cleanup between files.

Pseudonym Pools.

- **Person:** 32 gender-neutral English names (Alex, Jordan, Taylor, Morgan, Casey, Riley, Quinn, Avery, Parker, Drew, Reese, Jamie, Sage, River, Phoenix, Blake, Charlie, Emerson, Hayden, Skyler, Dakota, Finley, Rowan, Ellis, Cameron, Peyton, Logan, Spencer, Bailey, Kendall, Harper, Addison).
- **Location:** Coded patterns (Building-A, Room-101, Campus-North, City-A, Lab-Alpha, etc.).
- **Organization:** Coded patterns (Tech-Lab, Company-A, Institute-Alpha, etc.).

Quality Assurance. After automated PII removal, we ran a PII leak checker across all anonymized files (TXT, CSV, XLSX formats) to detect residual personal information. The Korean pipeline required many manual corrections, primarily due to common Korean names that are also everyday words (e.g., some names are identical to common pronouns).

F.4 Stage 4: Translation

To unify the dataset into a single language for downstream modeling, we translated all Korean transcripts to English using Qwen3-30B-A3B-Instruct-2507, running entirely on local machines.

All Korean transcripts were translated to English using Qwen3-30B-A3B-Instruct-2507 (bfloat16, device_map=auto). Translation was performed per-file in isolated subprocesses to ensure complete GPU memory release between files. Segments without Korean characters were automatically skipped via a has_korean() check. For segments exceeding 200 characters, individual (rather than batch) translation was used to maintain quality. Pseudonym preservation was validated post-translation using the validate_pseudonyms() function, which verifies that all pseudonyms present in the source appear in the translated output.

F.5 Stage 5: Cross-session Speaker Unification

Diarization assigns arbitrary speaker labels that reset across sessions, so the same individual may receive different labels in different recordings. We unified identities using WeSpeaker (Wang et al., 2023) 256-dimensional embeddings with the Hungarian algorithm (Kuhn, 1955) and a complementary *mention matrix*—leveraging the observation that speakers rarely say their own name—producing 422 speaker-file mappings with a within-speaker cosine similarity of 0.89 versus 0.34 between speakers.

Step 1: Embedding Extraction & Hungarian Matching. We used the WeSpeaker model (pyannote/wespeaker-voxceleb-resnet34-LM) to extract 256-dimensional speaker embed-

dings. For each speaker in each session, up to 5 segments were randomly sampled (minimum duration 1.5 s, capped at 30 s), and their embeddings were averaged to obtain a representative centroid. Files were processed in order of decreasing speaker count: the first file initializes the centroid pool, and subsequent files are matched against existing centroids using the Hungarian algorithm with cost matrix $C = 1 - \text{cosine_similarity}(E, M)$, where E is the embedding matrix and M is the centroid matrix. Unmatched speakers create new centroids (e.g., PERSON_F).

Step 2: Label Application. The resulting CSV mapping (422 entries) was applied to update the speaker field in all 105 JSON files, converting arbitrary per-session labels (PERSON_0, PERSON_1, ...) to consistent cross-session labels (PERSON_A, PERSON_B, ...). Team 1 was mapped manually and verified independently.

Step 3: Mention Matrix Inference. As a complementary identity signal, we counted how often each speaker mentions each pseudonym using word-boundary regex matching, leveraging the observation that speakers rarely say their own name. Each assignment received a confidence tag:

- **VERIFIED:** Manual verification (Team 1 only).
- **HIGH:** 0 self-mentions and ≥ 3 total mentions by others.
- **MED:** ≤ 1 self-mention and ≥ 2 total mentions.
- **LOW:** All other cases.

This produced 50 speaker-to-real-identity mappings across 12 teams.

F.6 Stage 6: Utterance Boundary Refinement

We refined utterance boundaries using GPT-5-mini via the OpenAI API, selected for its reliable structured JSON output and applied only to already-anonymized transcripts. The model decides one of four actions per utterance—KEEP, MERGE_NEXT, MERGE_PREV, or SPLIT—guided by 10 rules from the Language Development Project (LDP) transcription guidelines (MacWhinney, 2000), using a sliding window of 8 utterances. Hard constraints prevent merging across different speakers or pauses ≥ 2 s.

Model & Configuration. We used gpt-5-mini via the OpenAI API (AsyncOpenAI) with the following settings:

Parameter	Value
Model	gpt-5-mini
Window size	8 utterances
Stride	6 (overlap = 2)
Pause threshold	2.0 s
Max concurrent calls	10
Max retries	3 (exponential backoff)

Table 14: Utterance boundary refinement configuration.

LDP Rules. The system encodes 10 numbered rules from the Language Development Project transcription guidelines, plus 3 unnumbered heuristic rules (13 prompt items total). Two rules serve as hard constraints that cannot be overridden:

- **Rule 4.6:** Never merge utterances from different speakers.
- **Rule 4.8.1:** Never merge across pauses ≥ 2 seconds.

The remaining rules govern splitting and merging decisions:

Rule	Category	Action	Description
4.3	Self-correction	Keep	Within-thought corrections
4.4	Abandoned thought	Split	Different-thought restart
4.5	False start	Keep	Same topic, same utterance
4.7	Multi-sentence	Split	No conjunction between sentences
4.9.1	Non-sentence tag	Merge	"ok," "right," "honey"
4.9.3	Complete tag	Split	"I know," "I don't know"
4.9.4	Perception verb	Keep	"I know it's crazy" = 1 utt.
4.10	Repetition	Keep	"no no no" preserved

Table 15: LDP rules encoded in the utterance boundary refinement system.

821 **Processing Results.** Across 105 transcript files from 12 teams (89 unique meetings; longer
822 sessions were split into multiple parts):

- 823 • Splits applied: 1,238 (including multi-sentence splits producing 2–5 segments each);
- 824 Merges applied: 309
- 825 • API calls: 1,015 windows processed
- 826 • Error rate: 0% (all windows successful)

827 G Full Prediction Results

828 Table 16 shows the complete results for Experiment 1, including all proprietary models,
829 few-shot baselines, and Llama-3.1-8B fine-tuned conditions.

Model	Context	Type	Speaker		Intention	
			Acc.	Δ Bigram	Acc.	Δ Maj.
<i>Baselines</i>						
Random	—	—	30.22	—	7.14	—
Bigram	—	—	50.75	(ref)	—	—
Majority	—	—	—	—	30.05	(ref)
<i>Zero-shot / Vanilla</i>						
Llama-3.1-8B	SU	vanilla	29.07	−21.7	15.57	−14.5
Gemma-3-12B	SU	vanilla	43.56	−7.2	25.48	−4.6
GPT-5.4-mini	SU	zero-shot	59.23	+8.5	24.22	−5.8
Sonnet 4.6	SU	zero-shot	59.70	+9.0	25.38	−4.7
GPT-5.4	SU	zero-shot	61.97	+11.2	25.68	−4.4
Opus 4.6 [†]	SU	zero-shot	63.25	+12.5	26.68	−3.4
Gemma-3-12B	SU	7-shot	53.85	+3.1	—	—
<i>Fine-tuned (Gemma-3-12B-IT + LoRA)</i>						
Gemma-3-12B	SU	FT	64.53	+13.8	32.96	+2.9
Gemma-3-12B	SRU	FT	64.82	+14.1	32.53	+2.5
Gemma-3-12B	S+R	FT	65.74	+15.0	31.17	+1.1
<i>Fine-tuned (Llama-3.1-8B + LoRA)</i>						
Llama-3.1-8B	SU	FT	65.17	+14.4	33.18	+3.1
Llama-3.1-8B	SRU	FT	66.20	+15.5	32.82	+2.8
Llama-3.1-8B	S+R	FT	65.65	+14.9	31.72	+1.7

[†]Speaker: 154 API errors (3.0%); Intention: 57 errors (1.1%), treated as incorrect. Retrying yields 65.04% speaker accuracy; ranking vs. fine-tuned models is unchanged. Majority baseline for intention = "Giving Information" (30.05%). 7-shot intention not evaluated.

Table 16: Full prediction results for Experiment 1, including all models and conditions. See Table 1 for the main results.

H Model-Agnostic Validation

To confirm that our findings are not artifacts of a specific model, we replicate the full Experiment 1 conditions with Llama-3.1-8B-Instruct using identical LoRA configuration (rank 16, $\alpha = 32$, 2 epochs, lr = $2e-4$). Table 17 shows results across all six conditions.

Format	Mode	Speaker (%)		Intention (%)	
		Gemma	Llama	Gemma	Llama
SU	Vanilla	43.56	29.07	25.48	15.57
SU	FT	64.53	65.17	32.96	33.18
SRU	Vanilla	38.69	28.01	24.48	14.96
SRU	FT	64.82	66.20	32.53	32.82
S+R	Vanilla	47.78	42.64	21.50	11.17
S+R	FT	65.74	65.65	31.17	31.72

All FT results are within 0.1–1.4 pp across models for speaker prediction, and within 0.2–0.6 pp for intention prediction, suggesting model-agnostic patterns.

Table 17: Model-agnostic validation: Gemma-3-12B vs. Llama-3.1-8B across all conditions. Same LoRA config, same data splits.

I Learning Curve and AMI Experiment Details

Learning curve setup. For each team, we chronologically order all meetings and incrementally add them to the training set. At step k , the model trains on meetings 1 through k and is tested on all remaining meetings $k+1$ through N . This means that the test set shrinks as k increases, which is why late-stage accuracy estimates (e.g. Team 5 at 8–9 meetings) have high variance and occasionally exceed the 9-team ceiling. Training uses the same LoRA configuration as Experiment 1, with the SU (Speaker + Utterance) format. We additionally evaluate the 9-team general model (trained on all 32,243 samples) on each team’s full test data to establish a cross-team ceiling.

AMI data preparation. We use the AMI Meeting Corpus from HuggingFace (edinburghcstr/ami, IHM configuration). The official test split contains 16 meetings from 4 groups (EN2002, ES2004, IS1009, TS3003). We construct sliding-window samples with context-size 8 to match Hilgert & Niehues (2025), yielding 12,515 test samples. Speakers are anonymized to PERSON_A/B/C/D by order of first appearance in each meeting.

For the balanced TIDES+AMI training mix (124K), we use TIDES in unmerged format (62,221 samples, context 8) and downsample AMI training data from $\sim 108K$ to 62,221 samples to match. The unmerged format preserves finer utterance boundaries, that align with AMI’s segmentation style. The full unbalanced mix (170,038 samples, trained for 1 epoch) uses all available data from both corpora without downsampling, but degrades accuracy to 30.02%—even below the zero-shot baseline (35.33%)—suggesting that data balance matters more than volume.

J Training Composition and Cross-Culture Analysis

To understand how language composition in training data affects speaker prediction, we fix the test set (Teams 6&7, Korean) and vary training composition across four conditions: Korean-only, English-only, balanced (Korean downsampled to match English), and the original mixed set. Note that “Korean teams” and “English teams” refer to the language originally spoken during recorded meetings, not the actual language of the post-processed data.

Table 18 shows that **data volume is the primary driver**: Mixed (64.53%, 32K samples) outperforms all alternatives, and KR-only (60.21%) surpasses EN-only (57.20%) by 3.0 pp,

Condition	Lang.	Train N	Acc. (%)
Vanilla	—	0	43.56
Bigram	—	—	50.75
EN-only	100% EN	8,703	57.20
Balanced	50/50 KR/EN	17,406	58.74
KR-only	100% KR	23,540	60.21
Mixed (orig.)	73/27 KR/EN	32,243	64.53

Table 18: Effect of training language composition on speaker prediction. Test set fixed: Teams 6&7 (Korean). Gemma-3-12B-IT + LoRA.

reflecting a same-language advantage for the Korean test set. EN-only still exceeds the bigram baseline by +6.4 pp, suggesting that cross-lingual transfer of turn-taking dynamics is substantial—conversation structure transfers across languages even without target-language data. Balanced (58.74%) slightly exceeds EN-only but falls below KR-only, indicating that subsampling from 23K to 8.7K Korean samples carries a cost not fully offset by language diversity.

K Detailed Breakdowns

Per-role analysis. Speaker prediction accuracy varies by the target speaker’s emergent role. Attention Seekers are most predictable (75.4% on the test set), likely because their frequent backchanneling creates strong sequential patterns. Critics are hardest to predict (53.1%), consistent with their tendency to interject at unpredictable moments.

Tuckman stage progression. Teams in the Forming stage (Tuckman) show lower speaker prediction accuracy (57.8% on the test set) than teams in the Performing stage (70.3%), a +12.5 pp gap. This aligns with theory: established teams develop more predictable interaction patterns as they mature.

Intention class analysis. The fine-tuned model’s intention predictions concentrate on Giving Information (74%) and Active Listening (21%), achieving top-3 accuracy of 62.6% but macro F1 of only 0.072. This class collapse reflects both the skewed label distribution and the low human agreement ($\kappa = 0.339$), suggesting that finer-grained intention categories may require richer contextual signals or task-specific architectures.

L Generation Supplementary

Automatic evaluation. Table 19 shows that FT-B consistently outperforms FT-A on structural metrics (speaker accuracy, ROUGE-L) while V-A produces the most invalid speakers (786 at 10-turn vs. 152–178 for FT). Degeneration rates increase with generation length: 1% at 1-turn, 9% at 3-turn, and 23% at 5-turn, motivating the 4-stage quality filtering pipeline described below.

Surface post-processing. Manual inspection revealed that FT outputs, while contextually grounded, suffered from surface-level formatting issues absent in vanilla outputs, such as missing sentence-initial capitalization, missing punctuation, lowercase “i”, and truncated sentence endings. To ensure that evaluators judged content rather than formatting, we applied a two-pass surface polish to all FT utterances before evaluation. Pass 1 used GPT-5.4-mini to correct capitalization and basic punctuation (1,390/1,440 utterances modified). Pass 2 applied rule-based truncation fixes (278 trailing periods removed from incomplete sentences) followed by GPT-5.4-mini comma insertion (705 commas added). All changes were verified as surface-only. No words were added, removed, or reordered. Despite this polish, automated LLM judges still preferred vanilla outputs (83–87% across prompt variants), and the subsequent human evaluation confirmed the same pattern, indicating that the preference gap is not driven by surface formatting.

Condition	Speaker Acc. (%)			ROUGE-L (1t)	Inv. Spkrs [†]	Semantic Sim.		
	1t	3t	5t			1t	3t	5t
<i>Fine-tuned (Gemma-3-12B)</i>								
FT-A (Plain)	49.0	31.0	28.6	4.79	178	.307	.321	.345
FT-B (Social Cue)	57.0	37.7	29.2	5.46	152	.283	.317	.360
V-A (Vanilla)	14.0	21.7	20.2	4.41	786	.319	.346	.348
<i>Proprietary (zero-shot generation)</i>								
GPT-5.4	51.0	42.3	37.0	—	0	.364	.432	.441
GPT-5.4-mini	60.0	46.3	37.0	—	0	.360	.424	.436
Opus 4.6	62.0	46.3	39.0	—	0	.379	.411	.426
Sonnet 4.6	59.0	41.7	38.2	—	0	.396	.437	.440

[†]FT/V-A: counted at 10-turn generation; proprietary: per-turn (all 0). ROUGE-L not computed for proprietary (different prompt format).

Table 19: Automatic evaluation of generated utterances (100 samples per generation length). Speaker accuracy = fraction of correctly predicted speakers averaged across all generated turns. Semantic similarity = cosine similarity of sentence embeddings (all-MiniLM-L6-v2) between concatenated generated and ground-truth turns. **Bold** = best per column.

Quality filtering. We applied a 4-stage pipeline to select evaluation samples: (1) GPT-based defect filtering (5 defect categories, 3-run majority vote, 2,700 evaluations → 171 pass), (2) manual audit (3 removed), (3) subtle quality filtering (unnatural enumeration), and (4) uniform sampling (40 per generation length). Five-turn samples required regeneration with improved parameters (temperature 0.7→0.5, repetition penalty 1.2→1.3, best-of-3 selection). The final set comprises 120 samples × 3 pairs = 360 comparisons + 20 attention checks.

Human evaluation setup. We recruited evaluators through Prolific, requiring English fluency and prior experience with collaborative teamwork. The evaluation interface presented pairs of AI-generated meeting continuations (A and B, randomly assigned) alongside the original conversation context. Evaluators rated each pair on five dimensions: (1) overall preference (A much better / A slightly better / Tie / B slightly better / B much better), (2) confidence (1–5), and for each side separately, (3) naturalness (1–5), (4) coherence (1–5), and (5) speaker consistency (1–5).

Quality control used two mechanisms: attention checks (3 per evaluator, requiring identification of a clearly superior continuation) and instructed-response checks (requiring a specific Likert value). Evaluators failing ≥2 checks of the same type were terminated early. Of 123 total participants, 66 passed quality thresholds, yielding 1,142 approved responses across 360 comparison items (3 pairs × 120 samples).

L.1 LLM-as-Judge Evaluation

Table 20 compares two LLM judges. GPT-5.4 strongly aligns with human preferences (V-A preferred in 66.7–72.5% of comparisons involving V-A). Opus 4.6 diverges: it prefers FT-A over V-A overall (54.2% vs. 45.0%) and FT-B over V-A (62.5% vs. 37.5%), particularly at shorter generation lengths. At 1-turn, Opus prefers FT-A 72.5% of the time, but this reverses at 5-turn where V-A is preferred 67.5%. This suggests that Opus is more sensitive to structural coherence (where FT excels) whereas GPT prioritizes surface fluency (where V-A excels), highlighting that LLM-based evaluation of multi-party dialogue generation remains model-dependent.

Multi-turn fine-tuning failure. We attempted direct 5-turn fine-tuning (6,397 non-overlapping training samples), but 90% of generated outputs failed to parse correctly. We reverted to 1-turn fine-tuning with autoregressive multi-turn generation, which proved more reliable despite compounding errors over turns.

Context 5-turn continuation ▼ Hide

Harley: Yeah.

Riley: I thought about using camera to analyze the clothes but I'm not sure it's clever to analyze clothes thickness or color?

Pat: I think that would be too hard. like the camera itself. And then we would have to also code the camera too. So I think it's easier if we just hard-code it.

Harley: Yeah, I agree.

Pat: I think right now there's two many words in this page. Can we put the clothes data at the back?

Conversation A

Riley: Yeah. You can add more. The AI agent will ask questions based on user input here. It kind of sounds a bit off because how does your response change what you do today? Is it something different depending on where they live or weather conditions?

Pat: Right, yeah, okay, so maybe let me see the things I wrote down earlier. Yes, um, okay, so the persona has been built by analyzing various factors such as location, age, gender, occupation, personality traits, etc. While interacting with users, AI agents may adapt responses, tone.

Shiloh: Style conversation topics preferences from personal details inferred during onboarding process, providing personalized guidance, recommendations, tailored advice reflecting individual needs, interests, goals, and lifestyle choices, enabling meaningful engaging interactions between human and artificial intelligence, fostering strong relationships, trust over time, creating memorable

Conversation B

Shiloh: Good idea, B. It's getting a bit cluttered looking at all these different aspects crammed together. Moving the clothing details to later on will definitely improve readability for everyone reviewing this document. Let's do that - A.

Riley: Okay, sounds good. Yeah, you're right - my apologies for making things so dense! Happy to move those sections around; let me reorganize them and push everything related to clothing analysis towards the end of the document. Should

Harley: Just double checking though—when we do get to the clothing section at the end, should we still include some placeholder diagrams showing how we envision the user inputting their outfit choices? Even if it

Riley: ...even if the actual implementation detail isn't finalized yet? I was thinking maybe simple mockups could help illustrate the workflow regardless. Would that be useful, D?

Which is better?

A is much better

A is slightly better

Tie

B is slightly better

B is much better

Confidence

1 2 3 4 5

Low High

Rate each conversation

Naturalness

Does it sound like real people talking? Consider speech patterns, informal expressions, and natural conversational flow.

A 1 2 3 4 5

B 1 2 3 4 5

Coherence

Does the conversation logically flow from the context? Do the responses connect to and continue what was being discussed?

A 1 2 3 4 5

B 1 2 3 4 5

Consistency

Do speakers maintain their identity and stay true to their role without contradicting themselves?

A 1 2 3 4 5

B 1 2 3 4 5

Interestingness

Is the conversation engaging and content-rich, or generic and dull?

A 1 2 3 4 5

B 1 2 3 4 5

Submit & Next

Figure 3: Human evaluation interface on Prolific. Evaluators view the original conversation context (top), two AI-generated continuations A and B (bottom left/right, randomly assigned), and rate overall preference, confidence, naturalness, coherence, speaker consistency, and interestingness on the right panel.

Pair	GPT-5.4		Opus 4.6	
	A	B	A	B
FT-A vs FT-B	40.0%	60.0%	29.4%	51.3%
FT-A vs V-A	27.5%	72.5%	54.2%	45.0%
FT-B vs V-A	33.3%	66.7%	62.5%	37.5%

GPT-5.4 aligns with human evaluators (V-A preferred). Opus 4.6 diverges, preferring FT at shorter horizons.

Table 20: LLM-as-judge pairwise evaluation (360 comparisons each). A and B correspond to the first and second model in each pair. GPT-5.4 reports no ties; for Opus 4.6, remaining percentage = ties.

M Training and Inference Configuration

Table 21 summarizes the training hyperparameters for all fine-tuning experiments. All models use LoRA adapters with completion-only loss (prompt tokens masked). Inference for open-source models uses single-token logit scoring, where the prompt is fed through the model, and the candidate with the highest log-probability at the prediction position is selected. For proprietary models, we use greedy generative decoding (temperature 0, max tokens 50) via the respective APIs.

N Prompt Templates

We show the exact prompt format used for each task. Both open-source and proprietary models receive identical system and user messages.

Speaker prediction (SU format).

Parameter	Gemma-3-12B	Llama-3.1-8B
LoRA rank (r)	16	16
LoRA alpha (α)	32	32
LoRA dropout	0	0
LoRA targets	q, k, v, o, gate, up, down_proj	
Trainable params	$\sim 0.56\%$	$\sim 0.56\%$
Epochs	3 (2 for Mixed)	2
Learning rate	1e-4	2e-4
LR scheduler	cosine	cosine
Warmup ratio	0.1	0.1
Batch size	2	2
Gradient accum.	8	8
Effective batch	16	16
Max seq length	4,096	4,096
Precision	bf16	bf16
Optimizer	AdamW	AdamW
Hardware	2× NVIDIA RTX A6000 (48GB)	
Wall time (Exp 1)	$\sim 18\text{h}$	$\sim 8.5\text{h}$

Table 21: Training configuration for all fine-tuning experiments. All use LoRA with no quantization (full bf16).

945 **System:** You are an expert at predicting conversational dynamics in team meetings. Given
946 a dialogue history, predict which team member will speak next. Answer with only the
947 speaker ID (e.g., PERSON_A).
948 **User:** Below is a dialogue history from a team meeting. Predict which speaker speaks next.
949 Dialogue:
950 [Turn 1] PERSON_A: Hello. I'm Jordan.
951 [Turn 2] PERSON_B: I'm Alex.
952 [Turn 3] PERSON_C: I'm Drew.
953 [Turn 4] PERSON_D: I'm Reese.
954 [Turn 5] PERSON_A: It's being recorded, so just in case, I'll turn this one on too.
955 Participants in this meeting: PERSON_C, PERSON_B, PERSON_A, PERSON_D
956 Who speaks next? Answer with only the speaker ID.
957 **Expected output:** PERSON_B

958 Intention prediction (SU format).

959 **System:** You are an expert at predicting utterance types in team meeting conversations.
960 Given a dialogue history, predict the utterance type of the next turn.
961 [14 category definitions provided, e.g.:
962 - Active listening: Short utterances showing attention or agreement
963 - Giving Information: Providing objective facts or asking factual questions
964 ...]
965 Answer with only the utterance type name.
966 **User:** [Same dialogue format as speaker prediction, with utterance types annotated per turn,
967 e.g.:
968 [Turn 1] PERSON_A [Social / Humor]: Hello. I'm Jordan.]
969 **Expected output:** Active listening

970 **Generation (FT-A, plain format).** The model receives the 5-turn context and generates
971 the next utterance in "SPEAKER: utterance" format. For FT-B (social cue reasoning), the
972 model first generates a structured chain: "Next speaker: X / Role: Y / Intention: Z /
973 Utterance: text". For vanilla and proprietary models, the same context is provided, with
974 instructions to continue the conversation naturally.